DATA ALCOTT SYSTEMS • WEB DEVELOPMENT INTERNSHIP

CLAIM PRO

Insurance Claims Management System

A responsive, high-performance Single-Page Application (SPA) architected to streamline the full lifecycle of insurance claims—including multi-step intake filing, document verification, officer audit workflows, approval gating, financial settlement disbursement, and actuarial reporting.

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HOST ORGANIZATION
Data Alcott Systems
Reference: FreeInternships.in

INTERNSHIP IDENTIFIERS
Internship ID: 5696
Portal Task ID: #5110

PERIOD & DOMAIN
21 Oct 2026 – 28 Oct 2026
Domain: Web Development

CORE IMPLEMENTATION STACK (PURE CLIENT-SIDE ARCHITECTURE):
HTML5 Semantic Standards • CSS3 Custom Design System • Vanilla JavaScript (ES6+) • Browser LocalStorage Engine

GitHub: github.com/shresthchauhan14-cpu/claim-pro | Live Demo: shresthchauhan14-cpu.github.io/claim-pro

Introduction & Project Purpose

Modernizing insurance claim operations with a responsive, client-side digital workspace.

The Mission of CLAIM PRO

Eliminating Paper Inefficiencies

- Re-engineers the traditional insurance claims lifecycle into an intuitive, responsive Single-Page Application (SPA).
- Replaces cumbersome paper forms, physical bill submission, and disconnected spreadsheets with a unified digital surface.
- Provides total status transparency to policyholders while granting claims adjusters powerful auditing and verification tools.
- Eliminates communication blackouts by delivering real-time lifecycle tracking from submission to bank settlement.

End-to-End Operational Lifecycle

5 Cohesive Functional Phases

- 1. Guided Intake Filing: 4-step wizard with real-time validation and multi-file drag-and-drop document upload.
- 2. Document Verification Audit: Granular inspection of hospital bills, FIRs, and identity proofs with audit status toggles.
- 3. Officer Approval Gating: Intercepts approvals to prevent payouts if any document remains unverified or rejected.
- 4. Financial Settlement: Payout limit enforcement, transaction ID generation, and printable payment vouchers.
- 5. Actuarial Telemetry: Real-time SVG KPI trends, status distributions, and formula-injection-safe CSV exports.

Engineering Philosophy

Zero Framework & Backend Overhead

- Built strictly with pure modern web standards: semantic HTML5, custom tokenized CSS3, and vanilla ES6+ JavaScript.
- Operates entirely on client-side state persistence using browser LocalStorage with structured JSON schema serialization.
- Guarantees instant UI responsiveness and sub-second execution without external backend server latency.
- Zero installation barrier: Evaluators and users can execute and test the complete application instantly in any modern web browser.

Problem Statement & Industry Inefficiencies

Analyzing the systemic operational failures in legacy, paper-driven insurance claim management.

1. Severe Turnaround Latency & Tracking Gaps

Weeks Lost in Manual Paper Routing

- Physical claim forms and bills require inter-office postal dispatch, manual data entry, and multi-desk routing.
- Claimants face zero visibility into processing stages, overwhelming support staff with repetitive status inquiries.
- Missing or illegible documents stall claims indefinitely without automated claimant notifications.

2. Fraud Risks & Unverified Approvals

Lack of Compliance Gating & Verification Audits

- Physical bills and receipts are vulnerable to forgery, duplicate claims, and altered financial values.
- Legacy systems lack automated verification gates, allowing officers to approve claims before invoices are audited.
- No granular audit trails or permanent logs detailing which officer verified specific supporting documents.

3. Manual Settlement Calculation Errors

Overpayment & Reconciliation Discrepancies

- Manual payout calculation frequently leads to errors exceeding approved policy coverage caps.
- Lack of payment synchronization results in claims being marked settled before actual disbursement takes place.
- Absence of standardized, printable disbursement vouchers and verifiable transaction IDs.

4. Data Silos & Delayed Actuarial Insights

Lack of Real-Time Executive Telemetry

- Actuaries and executive leadership rely on fragmented spreadsheets compiled days or weeks after the period ends.
- No real-time dynamic KPI calculations for loss ratios, claim volumes, and approval percentages.
- Exporting client data into spreadsheets without sanitization introduces severe spreadsheet formula injection risks.

THE CORE CHALLENGE: Bridging the gap between speed, policyholder transparency, and strict underwriting compliance.

Core Project Objectives

Targeted architectural and functional deliverables established for the internship.

01

Objective 1: Responsive Single-Page Application (SPA)

Deliver an instantaneous, app-like desktop and mobile experience using clean client-side hash routing (#dashboard, #all-claims, #claim-review, #settlements, #documents, #reports, #settings) without page reloads.

02

Objective 2: Guided 4-Step Validated Intake Wizard

Implement a streamlined claim intake workflow with defensive client-side validation—enforcing 10-digit phone checking, positive numeric amounts, future date blocking, and drag-and-drop document upload with format checking.

03

Objective 3: Strict Document-Verification Compliance Gate

Enforce core underwriting compliance where a claim strictly cannot be approved if any attached supporting document is unverified or marked rejected. Clicking approve intercepts the action and shows a compliance modal.

04

Objective 4: Automated Settlement & Disbursement Lifecycle

Build a financial disbursement module that enforces payout ceilings (disbursements cannot exceed approved benefit), synchronizes terminal status ('Settled' only when 'Paid'), and generates printable vouchers.

05

Objective 5: Real-Time Actuarial Telemetry & Safe CSV Export

Calculate all 10 dashboard KPIs, inline SVG trend curves, and category donut charts directly from underlying data on every mutation, paired with RFC-compliant CSV export secured against spreadsheet formula injection.

05 | TECHNICAL FOUNDATION

Technology Stack & Implementation Roles

Pure modern web standards chosen for performance, accessibility, and zero-dependency execution.

HTML5 Semantics

Structure & Native Modals

- Semantic elements (<header>, <nav>, <aside>, <main>, <dialog>).
- Native <dialog> elements with keyboard ESC closing listeners.
- ARIA attributes for screen readers and accessibility.
- Accessible form controls with dynamic validation visual states.
- Zero third-party icon fonts or heavy CDN dependencies.

CSS3 Design System

Tokens, Layouts & Theming

- Custom design tokens (--primary: #7C3AED, --bg-app: #F8FAFC).
- Advanced CSS Grid and Flexbox for fluid layouts (320px–1440px).
- High-contrast Dark Mode toggling via data-theme='dark'.
- Dedicated @media print stylesheets for payment vouchers.
- Zero CSS frameworks (no Tailwind or Bootstrap bloat).

Vanilla JavaScript

State & Business Logic

- Centralized Object-Oriented Store architecture (ClaimStore).
- Event delegation and asynchronous DOM rendering updates.
- Blob URL creation (URL.createObjectURL) for session file previews.
- Spreadsheet Formula Injection defense for CSV export.
- Zero build steps, bundlers, or heavy external web frameworks.

Browser LocalStorage

Client Persistence Engine

- JSON-serialized state persistence across browser reloads.
- Zero server dependency for evaluator ease of testing.
- Pre-seeded realistic demonstration data across all categories.
- Single-click factory data reset and JSON backup/restore.
- Defensive try/catch parsing to guard against storage corruption.

Frontend System Architecture & Data Flow

Decoupled, modular client-side architecture operating on browser LocalStorage.

Layer 1: Presentation & UI Layer

Executive Dashboard • 4-Step Wizard • Master Claims Directory • Officer Review Queue • Settlement Manager • Reports & Modals

Layer 2: Controller & Routing Layer

SPA Hash Router (#hash) • Form Validation Engine • Dynamic DOM Renderers • Math-Driven Inline SVG Chart Calculators

Layer 3: Central Business Logic Store (ClaimStore)

Document-Verification Approval Gate • Payout Ceiling Enforcement • Spreadsheet Formula Sanitizer • Role Switcher (RBAC)

Layer 4: Client Persistence Engine

Browser LocalStorage (JSON Engine) • Temporary Session Blob URLs (URL.createObjectURL) • Factory Reset & Backup Importer

CRITICAL ARCHITECTURAL SPECIFICATION: PURE FRONTEND DEMONSTRATION

CLAIM PRO is strictly implemented as a client-side web application leveraging browser LocalStorage for all state persistence. It does NOT utilize a production backend server, external microservice, or cloud database. This allows evaluators and users to run and test the complete system instantly without server setup or network dependencies.

Comprehensive Application Feature Matrix

Six core functional modules verified in source code and end-to-end testing.

1. Executive KPI Dashboard & Telemetry

10 real-time metric cards (Total Claims, Settled, Under Review, Approval Rate, ₹19.18L Volume), dynamic SVG filing trend line, category donut chart, and live audit feed.

2. 4-Step Claim Intake Wizard

Multi-stage intake wizard with dynamic validation, Indian 10-digit mobile check, positive amount validation, future incident date blocking, and drag-and-drop file upload.

3. Master Claims Directory & Archive

Searchable, filterable, sortable master archive with multi-parameter matching, status badge indicators, and client-side pagination across all claim categories.

4. Officer Review & Document Audit Queue

Granular document inspection (hospital bills, FIRs, IDs) with individual status toggles (Verified/Pending/Rejected), audit notes logging, and approval lockout guards.

5. Settlement & Disbursement Management

Payout limit validation (disbursement cannot exceed approved claim), payment method selection, transaction ID assignment, and printable payment vouchers.

6. Actuarial Reports & CSV Export

Lifecycle distribution progress bars, insurance category financial breakdown matrices, and secure RFC-compliant CSV data export with formula injection defense.

Executive Dashboard & Real-Time Analytics

Dynamic business intelligence with zero third-party chart dependencies.

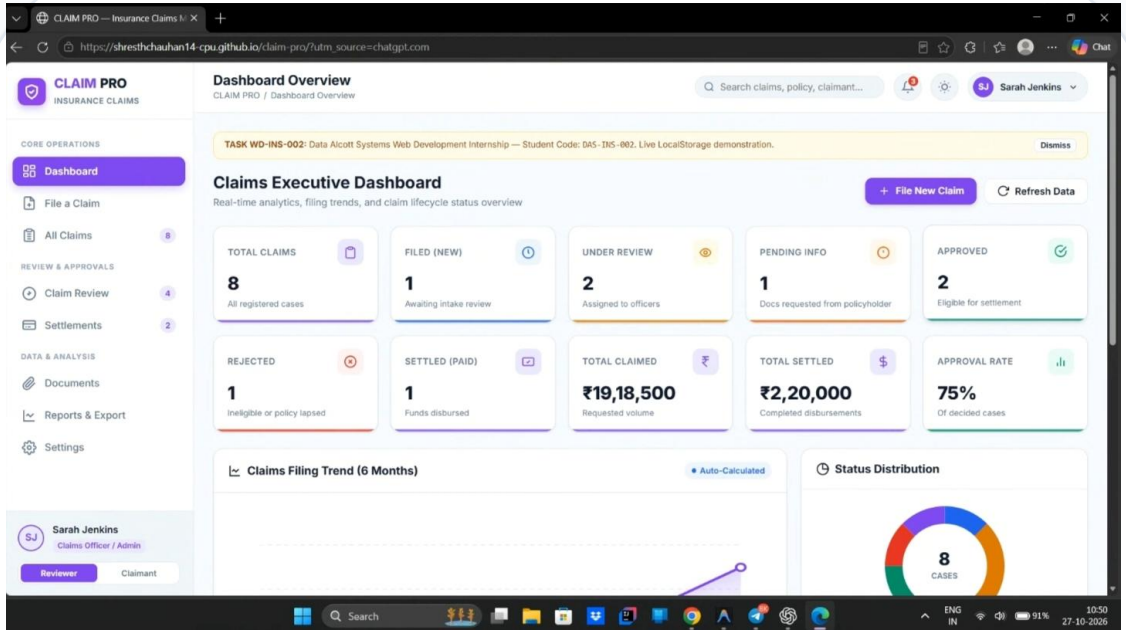


FIGURE A1: CLAIMS EXECUTIVE DASHBOARD & ANALYTICS OVERVIEW
Actual desktop screenshot displaying 10 calculated KPI cards, inline SVG 6-month filing volume trend line, interactive category donut chart, and live activity audit log.

- **10 Dynamic KPI Metric Cards**
  Calculates Total Claims (8), Under Review, Approved, Settled, Total Claimed Volume (₹19,18,500), and Approval Rate (62.5%) directly from state upon every mutation.

- **Math-Driven Inline SVG Trend Chart**
  6-month historical claim volume trajectory plotted dynamically using SVG <polyline> and math coordinates without external charting libraries (Chart.js/D3).

- **Dynamic Category Donut Chart**
  Calculates SVG circle stroke-dasharray segments in real time, breaking down volume across Health, Auto, Life, and Home insurance lines.

- **Live System Activity Audit Log**
  Chronological feed tracking recent claim submissions, officer approvals, document verifications, and financial disbursements in real time.

Guided Multi-Step Claim Intake Wizard

Defensive input validation and drag-and-drop document ingestion.

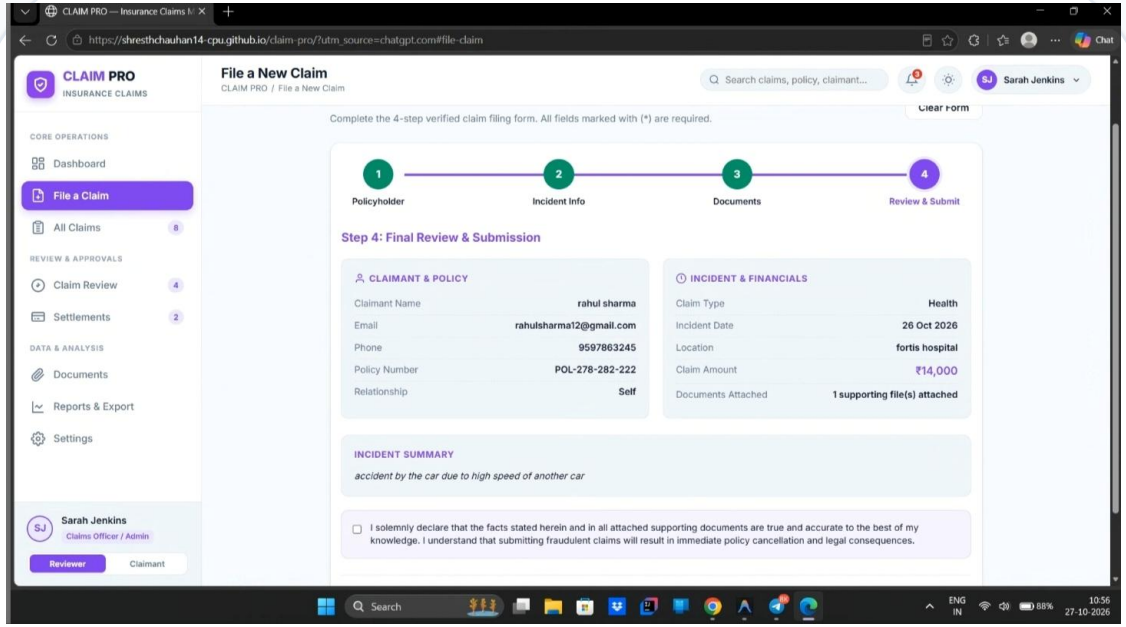


FIGURE A17: MULTI-STEP CLAIM INTAKE — STEP 4: FINAL REVIEW

Actual screenshot of the guided 4-step wizard displaying the final summary confirmation modal prior to dispatching into the central Store and generating unique claim IDs.

Step 1: Claimant & Policyholder Details

Captures claimant name, policyholder identity, email, and enforces strict 10-digit Indian phone number formatting with real-time regex validation.

Step 2: Policy & Coverage Selection

Validates policy number format and selects insurance category (Health, Auto, Life, Property) with pre-filled benefit limit parameters.

Step 3: Incident Details & File Upload

Blocks future incident dates via date picker constraint; positive amount validator; drag-and-drop file uploader accepting PDF, PNG, JPG with file size guards.

Step 4: Final Review & Confirmation

Consolidated review modal allowing policyholders to verify all details before dispatching into the central Store, generating a unique claim ID (CLM-2026-XXXX).

Officer Review & Document Verification Gate

Enforcing compliance rules to prevent premature claim approvals.

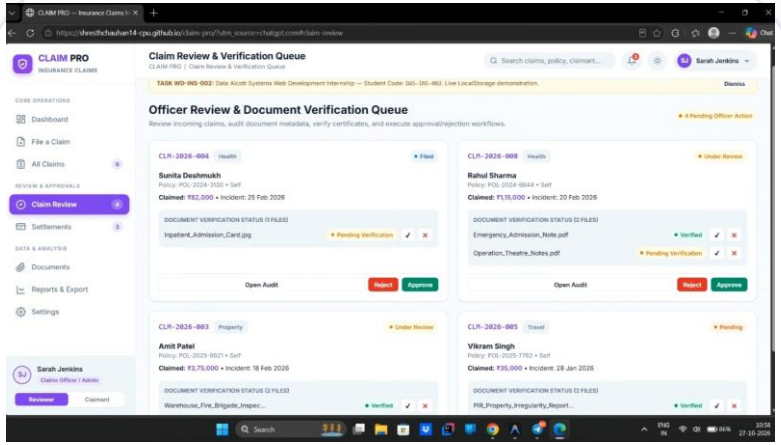


FIGURE A13: OFFICER REVIEW QUEUE
Granular officer queue displaying attached document status badges (Verified, Pending, Rejected) and review controls.

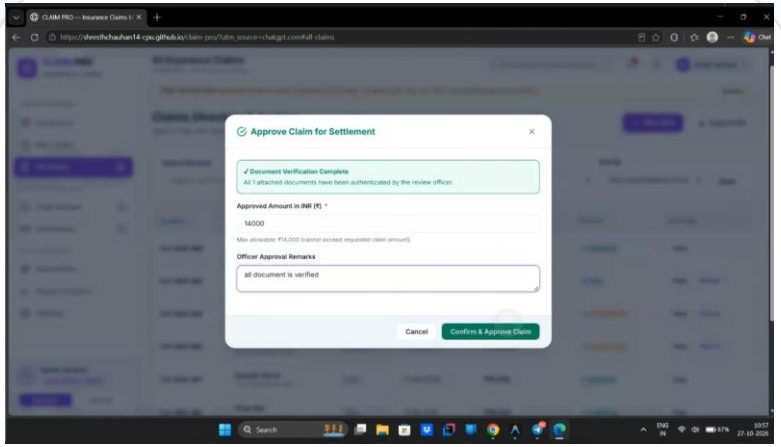


FIGURE A15: SETTLEMENT APPROVAL GATE
Compliance guard intercepting approval and detailing mandatory document verifications before status can transition to Approved.

- **Granular Document Audit**
  Claims officers inspect medical records, police FIRs, and bills with dedicated status toggles: Verified, Pending, Rejected.

- **Approval Lockout Guard**
  A claim cannot transition to Approved if any document remains Pending or Rejected. The system blocks invalid transitions.

- **Rejection Reason Logging**
  Rejecting a claim mandates an explicit reason string, permanently recorded into the claim's chronological timeline.

- **Complete Audit Trail**
  All officer actions log timestamped milestone entries into the claim's permanent audit history.

Settlement Processing & Actuarial Reports

Automated payout validation and secure data export.

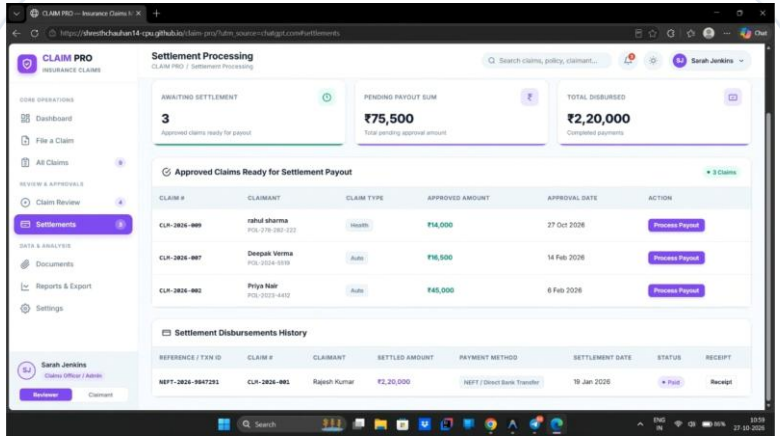


FIGURE A12: SETTLEMENT MANAGEMENT

Approved claim settlement processing with payout ceiling validation, payment channel selection, and voucher triggers.

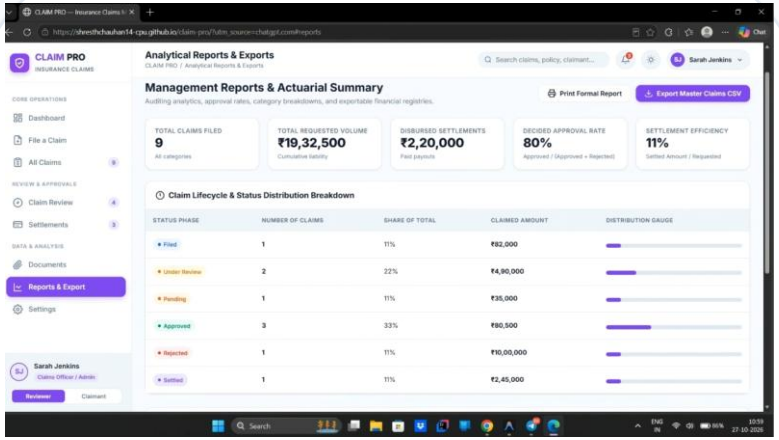


FIGURE A10: ACTUARIAL REPORTS

Desktop analytical breakdown displaying category distribution bars, financial aggregates, and CSV data export modules.

- **Payout Ceiling Enforcement**
  Disbursement amounts are validated to ensure they are strictly positive and do not exceed the approved claim benefit cap.
- **State Synchronization**
  A claim is only marked Settled once payment status is confirmed as Paid. If Processing, it remains in the Approved state.
- **Printable Vouchers**
  Dedicated @media print stylesheet generates bank-ready payment vouchers complete with transaction IDs and policy metadata.
- **Spreadsheet Formula Defense**
  CSV exports sanitize cells starting with '=', '+', '-', '@' by prepending a single quote, neutralizing Excel command injection.

Responsive Architecture & Mobile Optimization

Seamless user experience across smartphones, tablets, and desktop workstations.

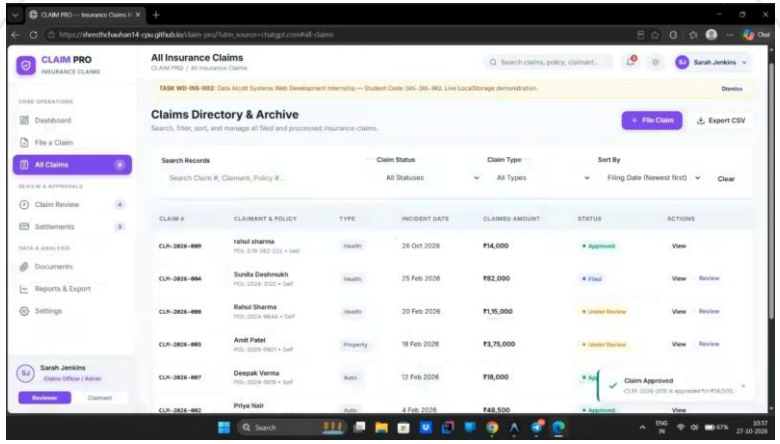
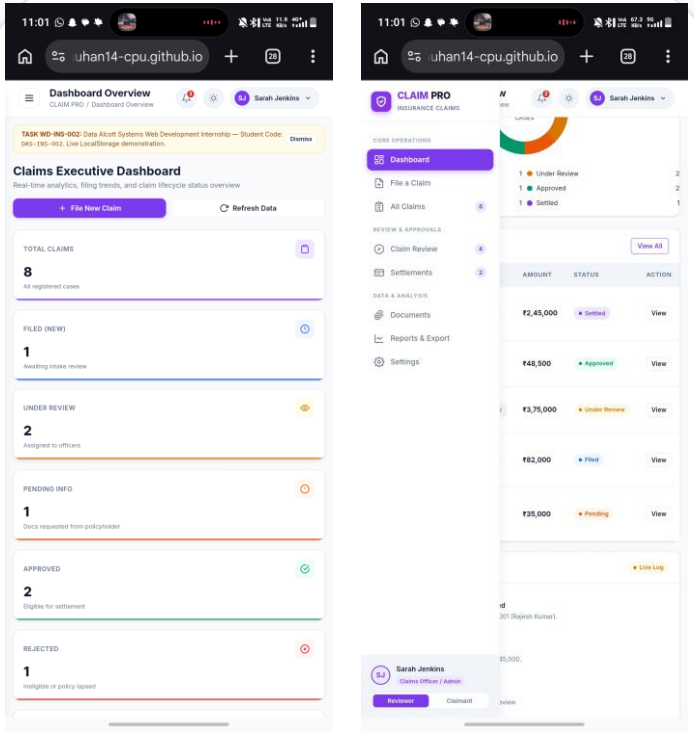


FIGURE A14: DESKTOP CLAIMS ARCHIVE
Desktop layout showing persistent sidebar, search bar, sortable claims table, and interactive toast notification.



FIGURES A2 & A5: MOBILE DASHBOARD & OFF-CANVAS DRAWER

- **Mobile-First Breakpoints**
  Custom media queries engineered from 320px mobile screens to 1440px ultra-wide monitors.
- **Off-Canvas Navigation Drawer**
  Desktop persistent sidebar gracefully converts to a sliding off-canvas drawer on mobile screens with backdrop touch dismissal.
- **Touch Target Optimization**
  All interactive buttons and table controls meet 48px minimum touch target standards.
- **High-Contrast Dark Mode**
  Instant dark theme toggling tokenized with CSS variables, preserving user preference in LocalStorage.

Technical Challenges & Key Learnings

Real-world engineering hurdles overcome during the internship sprint.

Challenge 1: LocalStorage 5MB Quota Limits

Managing Binary Document Uploads

Problem: Storing uploaded PDF/image document binaries as base64 strings quickly exceeds the browser's 5MB LocalStorage quota.

Solution: Stored document metadata (name, size, type, upload date, verification status) in LocalStorage while generating temporary URL.createObjectURL references for in-session previews.

Challenge 2: Multi-Stage Workflow Rules

Enforcing Status Integrity Without Backend

Problem: In a client-only app without server authorization, users could theoretically approve claims before required incident bills were audited.

Solution: Designed an Object-Oriented Store controller that intercepts state transitions and enforces prerequisite guards (e.g., verifying all documents before approval).

Challenge 3: Client Data Export Security

Spreadsheet Formula Injection Defense

Problem: CSV exports containing user-entered claim descriptions starting with '=', '+', '-', or '@' could execute arbitrary shell commands when opened in Excel.

Solution: Implemented RFC-compliant quoting and defensive sanitization by prepending a single quote (') to threat prefixes, neutralizing execution.

CORE LEARNING OUTCOMES: Mastered modular vanilla JavaScript state architecture, defensive CSV data sanitization, dynamic inline SVG coordinate calculation, and accessible UI component engineering.

Conclusion & Future Scope

Summary of completed project outcomes and roadmap for enterprise production.

Completed Internship Deliverables (Verified)

✓ Full Lifecycle SPA Completed

Engineered and verified a complete, responsive Single-Page Insurance Claims Management System within the 1-week internship period.

✓ 52 Automated Tests Passing

Validated with 52 automated Node.js test suites with 0 syntax errors, robust formula injection defense, and input validation checks.

✓ Google Chrome Headless Verification

Verified DOM rendering, dynamic SVG chart calculations, and client-side hash routing using Chrome CLI (--headless --dump-dom).

✓ Defensive Programming Proven

Demonstrated that vanilla web standards can achieve enterprise-grade UI fluidity, validation rigor, and accessibility without heavy external frameworks.

Future Scope & Production Roadmap (Planned)

→ Production REST/GraphQL API Backend

Transition from LocalStorage to a high-throughput backend service built with Node.js/Express or Python FastAPI with rate limiting and logging.

→ Relational Database (PostgreSQL)

Replace browser storage with PostgreSQL 16 featuring ACID transactions, foreign key constraints, and multi-tenant Row Level Security (RLS).

→ Encrypted Cloud Document Storage

Store original PDF and image bills in AWS S3 or Google Cloud Storage buckets with signed URL access control meeting legal compliance.

→ Server-Side Authentication & RBAC

Implement OAuth2 / JWT authentication with cryptographic session cookies and server-enforced role access controls.

→ AI/OCR Automated Bill Verification

Integrate optical character recognition (OCR) to automatically extract bill amounts and hospital IDs from uploaded images.

Thank You!

CLAIM PRO — Insurance Claims Management System | Web Development Internship

VERIFIED PROJECT ARTIFACTS & EVALUATION LINKS:

Live Interactive Demo: https://shresthchauhan14-cpu.github.io/claim-pro/

Source Code Repository: https://github.com/shresthchauhan14-cpu/claim-pro

YouTube Walkthrough Video: https://www.youtube.com/watch?v=Gdbz637qS1E

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**Note on Certification: Official internship completion certificates will be appended to the documentation repository upon issuance by Data Alcott Systems.*